

Task 1 – Laptop will not power on

A laptop that does not power on should be checked from the simplest external causes toward internal hardware faults. The following sequence helps avoid replacing a component before the basic power path has been confirmed.

1. Check the wall socket and adapter – First, connect another known-working device to the same socket. Inspect the laptop charger, plug and cable for cuts, bent pins or loose connections. This checks whether the problem is caused by the external power source rather than the laptop.
2. Check the charging indicator – Connect the charger to the laptop and observe the charging LED. If the indicator does not appear, the adapter, DC-in jack, battery or charging circuit may be involved. If the light appears but the laptop remains completely dead, continue with the next checks.
3. Perform a basic power reset – Disconnect the charger and, if the battery is removable, remove it. Hold the power button for about 15–20 seconds to discharge residual power. Reconnect the charger and try starting the laptop again. This can clear a temporary power-state problem.
4. Test with the battery and adapter separately – If possible, try starting the laptop with the battery removed and the AC adapter connected. If it works on AC power but not from the battery, the battery may be weak or faulty. If it does not work on either source, the fault may be in the charging jack, power circuit or motherboard.
5. Look for signs of life – Check for fan movement, keyboard LEDs, charging lights or display activity. If there are lights or fan activity but no picture, the problem may actually be a display, RAM or POST-related fault rather than a complete power failure.

Result: These checks narrow the fault from the wall supply and charger to the battery and finally to the laptop's internal power circuitry. The laptop should not be opened until the external power checks have been completed safely.

Task 2 – Monitor shows a blank screen

When the CPU appears to be running but the monitor is blank, the troubleshooting should separate the display path from the computer itself.

1. Confirm that the monitor is powered – Check the monitor's power LED and power cable. Make sure the monitor is switched on and not simply in standby mode.
2. Check the selected input – Use the monitor's input/source button and select the port actually being used, such as HDMI, DisplayPort or VGA. A wrong input selection can produce a blank screen even when the PC is working normally.
3. Reseat the video cable – Turn off the system if necessary and reconnect the display cable at both ends. Look for loose connectors or damaged pins. If available, test with another known-good cable.
4. Check the correct video output – If the computer has a dedicated graphics card, the monitor cable normally needs to be connected to the graphics card's output rather than an unused motherboard video connector. This is especially important when the processor does not provide integrated graphics.
5. Test another display or port – Connect the computer to another monitor or TV, or use another video output on the system. If the second display works, the original monitor, cable or input may be the problem.
6. Check hardware connections – If there is still no display, power off the PC and, after following safe anti-static precautions, check whether RAM and the graphics card are seated correctly. A poor connection can stop the system from completing POST.
7. Check basic display settings – If the display appears after Windows starts but later becomes blank, try Windows display shortcuts such as Win + P and select the appropriate display mode. A software or display-mode issue can look similar to a hardware fault.

Result: The fault is isolated by testing the monitor, input, cable, video output and internal hardware one at a time instead of changing several things simultaneously.

Task 3 – Wireless mouse is unresponsive

Because the keyboard works, the computer itself is probably receiving at least some input. The aim is to determine whether the problem is with the mouse, its USB receiver/connection, the USB port or system settings.

1. Check the mouse power – For a wireless mouse, replace or recharge the batteries and check the power switch. Look for an optical sensor light if the model provides one.
2. Check the receiver – If the mouse uses a USB receiver, remove it and connect it directly to another USB port. Avoid using a loose USB hub during testing.
3. Test the USB port – Connect another known-working USB device to the same port. If that device also fails, the port or its connection may be the problem.
4. Test the mouse on another computer – If the mouse fails on another computer as well, the mouse or its receiver is likely faulty. If it works elsewhere, the original computer's USB or software configuration should be investigated.
5. Re-pair the mouse if required – Some wireless mice use a pairing button or manufacturer utility. Follow the normal pairing process and confirm that the receiver and mouse are communicating.
6. Check system settings – In Windows, inspect Device Manager for mouse/HID or USB-related warning symbols. Reconnecting the device or restarting the system can also allow Windows to reload the device.
7. Confirm the final result – Move the mouse and test the left/right buttons and scroll wheel. A successful test should confirm stable pointer movement and normal button operation.

Result: Testing another USB port, another device and another computer isolates the fault efficiently without immediately assuming that the mouse itself is defective.

Task 4 – Gaming PC overheats and shuts down

Unexpected shutdown during gaming can occur when the CPU or GPU reaches a protective temperature limit. Three common hardware causes are restricted airflow, a faulty cooling fan, and degraded or incorrectly applied thermal paste.

Cause 1: Dust and blocked airflow Dust can collect in the front filters, heatsinks, fans and exhaust areas. This restricts airflow and causes hot air to remain inside the case. To check it, power down the PC, disconnect it from mains power and inspect the intake filters, CPU heatsink, GPU heatsink and exhaust fans. Heavy dust buildup is a strong indication that cleaning is required.

Cause 2: Fan or cooling-pump problem A CPU or case fan that is stopped, noisy or spinning slowly may not move enough air. During a controlled test, observe whether the relevant fans start normally. A liquid-cooling system may also have a pump or radiator-fan problem. A failed cooling component should be replaced rather than operated under heavy load.

Cause 3: Thermal paste or heatsink contact Thermal paste transfers heat from the processor to the heatsink. If the paste has dried out, was applied incorrectly, or the heatsink is not making proper contact, heat transfer becomes inefficient. The heatsink mounting should be checked for secure, even contact. If required, the old paste can be cleaned and fresh suitable thermal paste applied.

Additional checks: - Make sure the case has a clear intake and exhaust path. - Confirm that fans are installed in the correct airflow direction. - Check whether the shutdown happens only during heavy gaming load. - Test the system after cleaning and cooling maintenance. - If overheating continues, inspect the CPU cooler, GPU cooling system and power delivery.

Result: The safest approach is to identify the heat source, inspect the cooling path, correct the physical problem, and then confirm stable operation under normal load. Repeated forced shutdowns should not be ignored because excessive heat can shorten component life.